

EC571 Lab 2

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1 Problem 1: Voltage Transfer Characteristic (VTC)

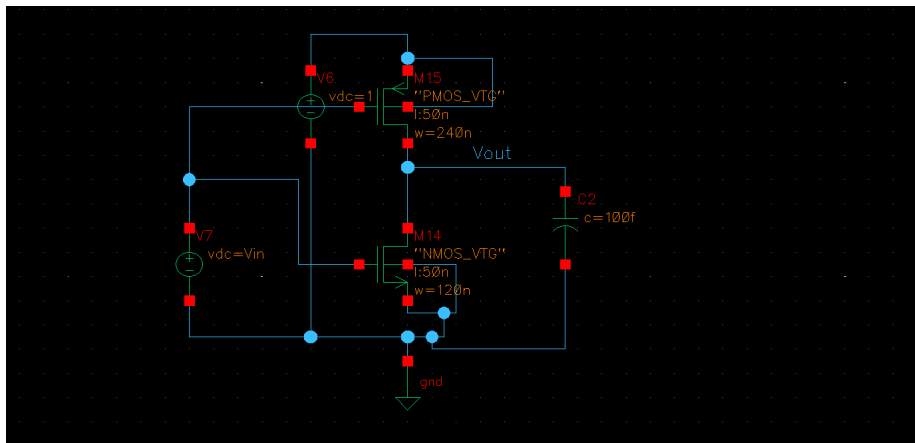


Figure 1: Inverter

1.1 Part a

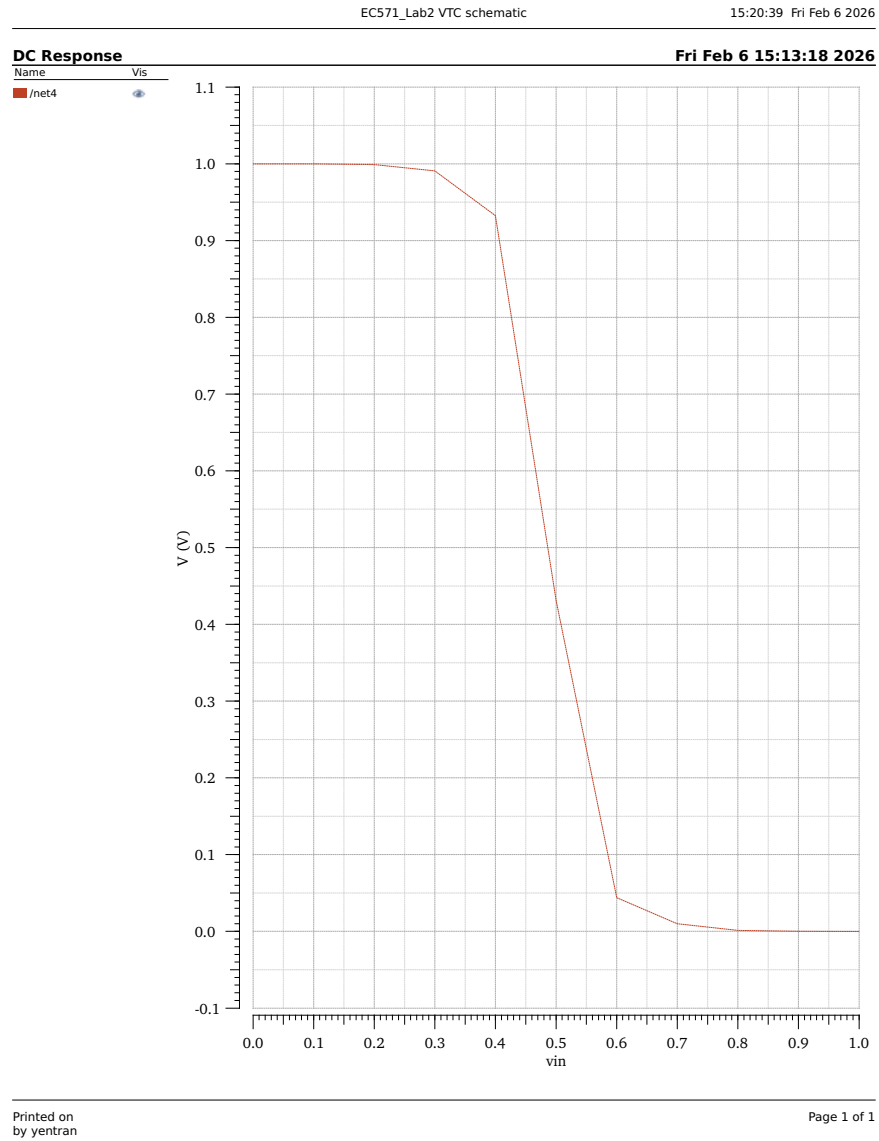


Figure 2: VTC of Inverter with $\frac{w_p}{w_n} = \frac{120nm}{120nm} = 1$ and $l_p = l_n = 50nm$

1.2 Part b

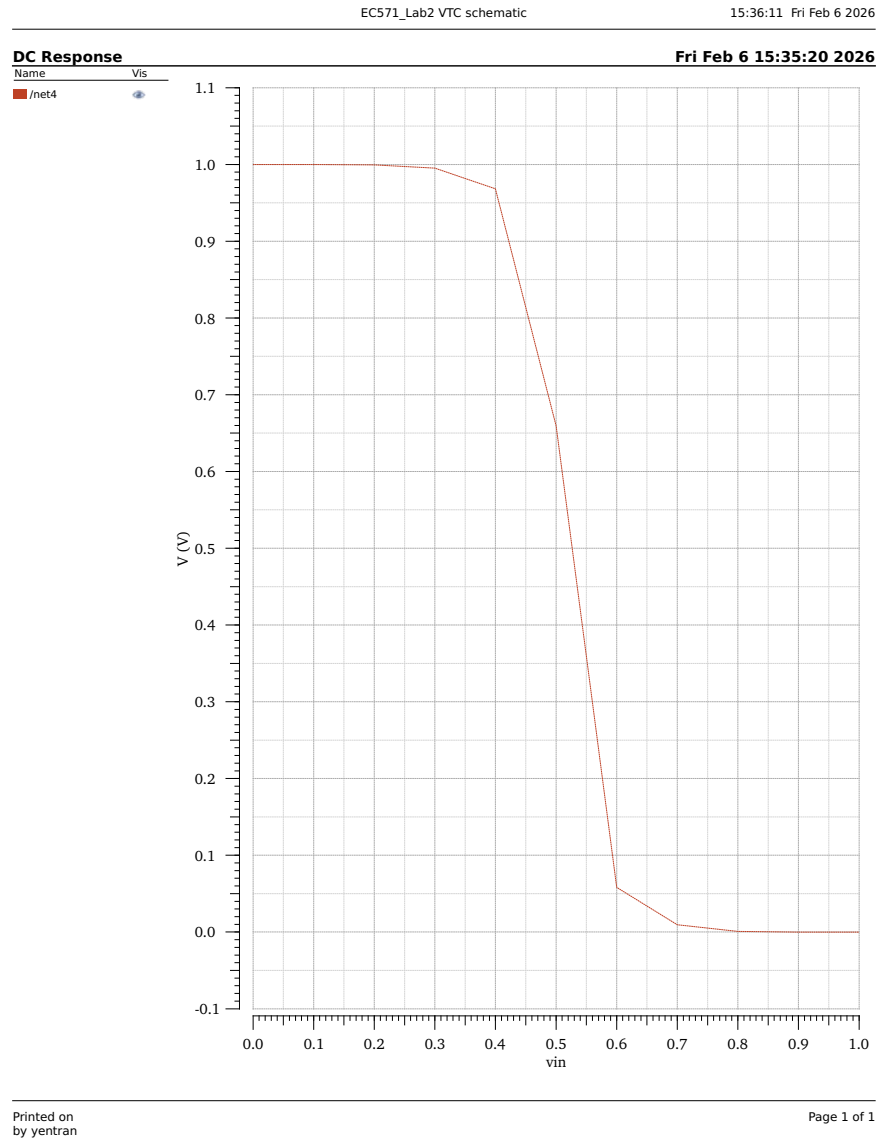


Figure 3: VTC of Inverter with $\frac{wp}{wn} = \frac{240nm}{120nm} = 2$ and $lp = ln = 50nm$

1.3 Part c

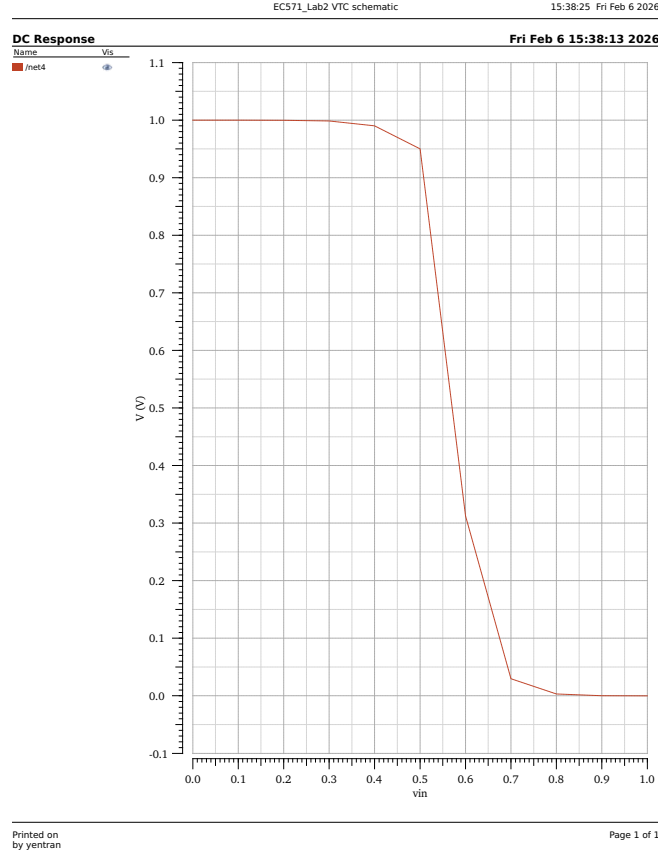


Figure 4: VTC of Inverter with $\frac{w_p}{w_n} = \frac{720nm}{120nm} = 6$ and $l_p = l_n = 50nm$

1.4 Part d

The Inverter with $\frac{w_p}{w_n} = 2$ has a relatively symmetrical noise margin. Because $\frac{\mu_n}{\mu_p} = 2$. The higher ratio between pmos to nmos, the more skewed the switching point (V_m) towards to the right. The Inverter with $\frac{w_p}{w_n} = 6$ has switching point skewed most significantly to the right shrinking the noise margin high region (NM_H) while noise margin low region (NM_L) expand. The Inverter with $\frac{w_p}{w_n} = 1$ has stronger nmos, shifting switching point (V_m) more skewed to the left, shrinking the noise margin low region(NM_L), while expanding noise margin high region (NM_H). Among the inverters, the inverter with $\frac{w_p}{w_n} = 1$ has the largest Noise Margin High region(NM_H).

2 Problem 2: Transient Response

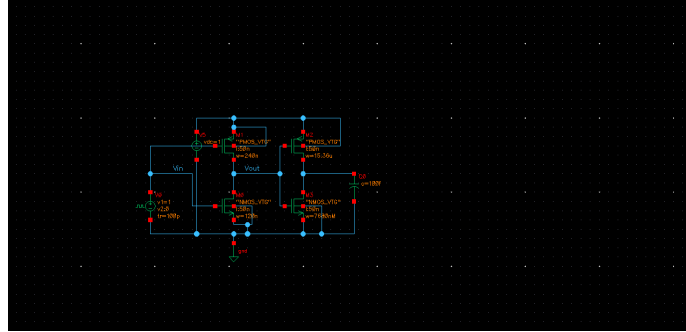


Figure 5: Two inverters

2.1 Part a

2.1.1 Propagation delay

$$t_{pHL} = 14.78\text{ps}$$

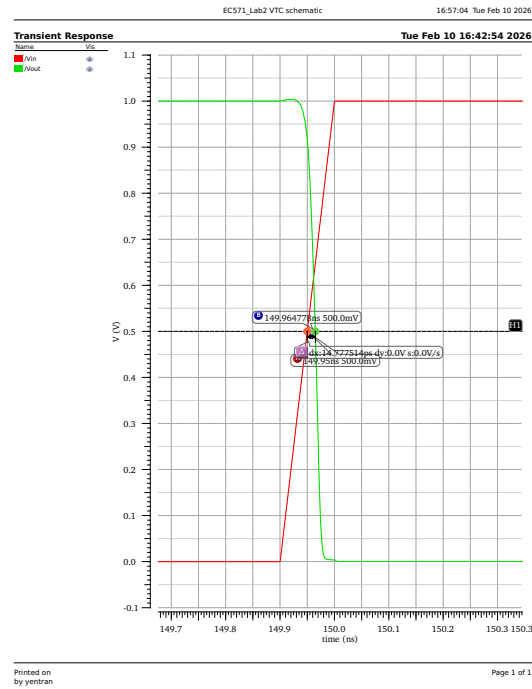


Figure 6: Propagation delay from High to Low (t_{pHL})

$$t_{pLH} = 11.19\text{ps}$$

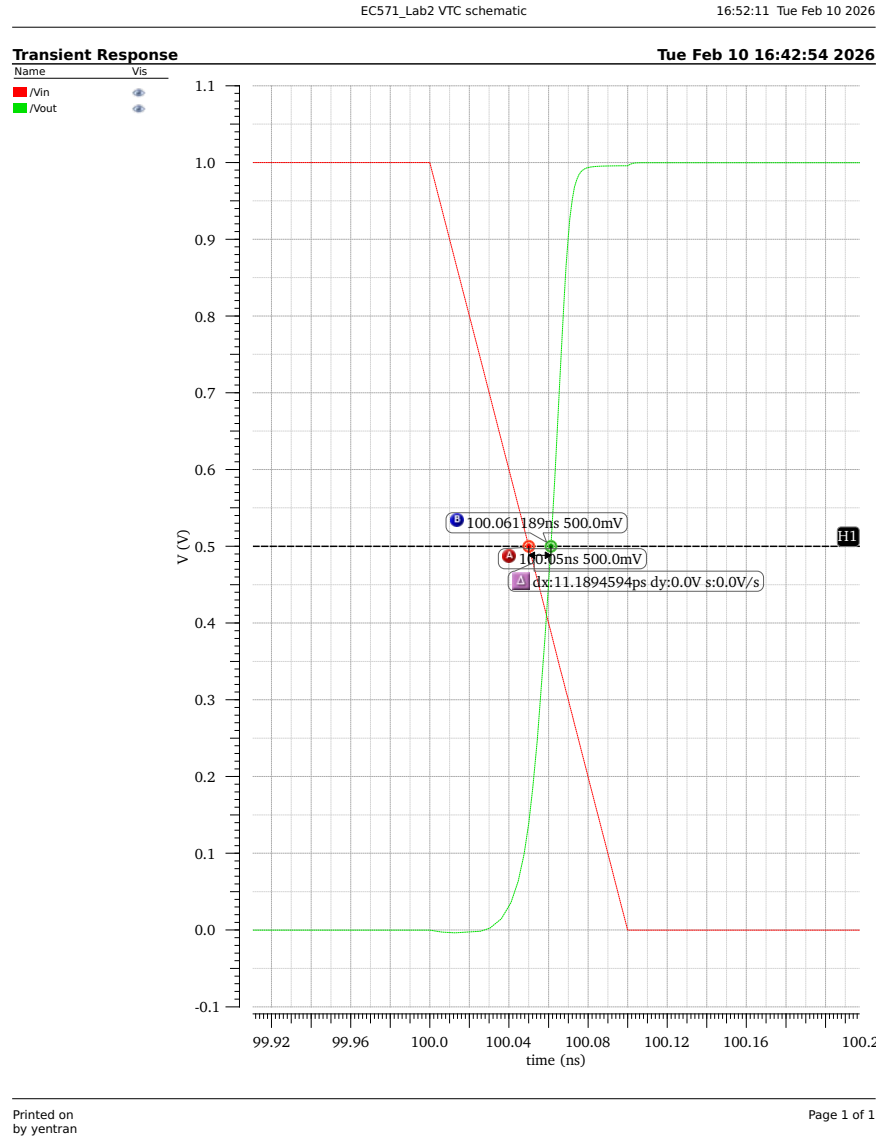


Figure 7: Propagation delay from Low to High (t_{pLH})

$$\text{Propagation delay}(t_p) = \frac{t_{pHL} + t_{pLH}}{2} = \frac{14.78 \text{ ps} + 11.19 \text{ ps}}{2} = 12.99 \text{ ps}$$

2.1.2 Rise time and Fall time

Rise time (t_r) = 22.15 ps

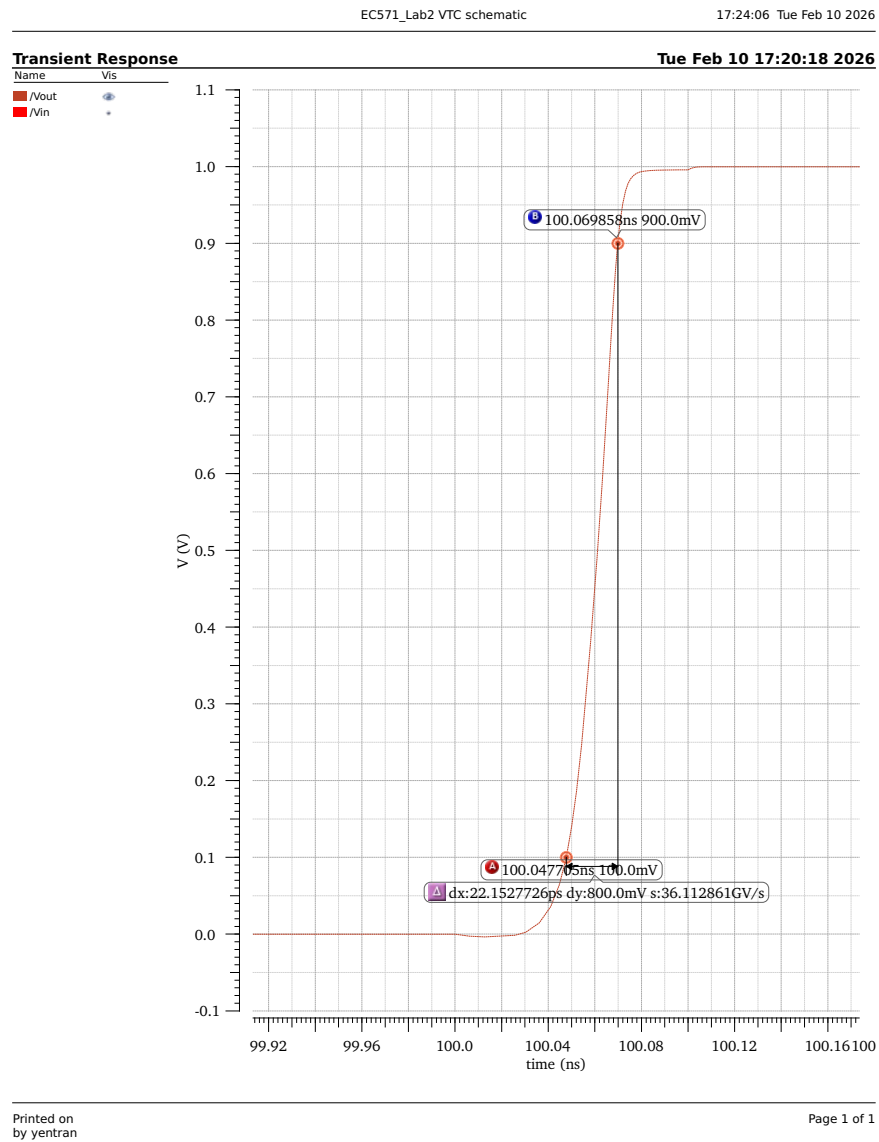


Figure 8: Rise time (t_r)

Fall time (t_f) = 23.03 ps

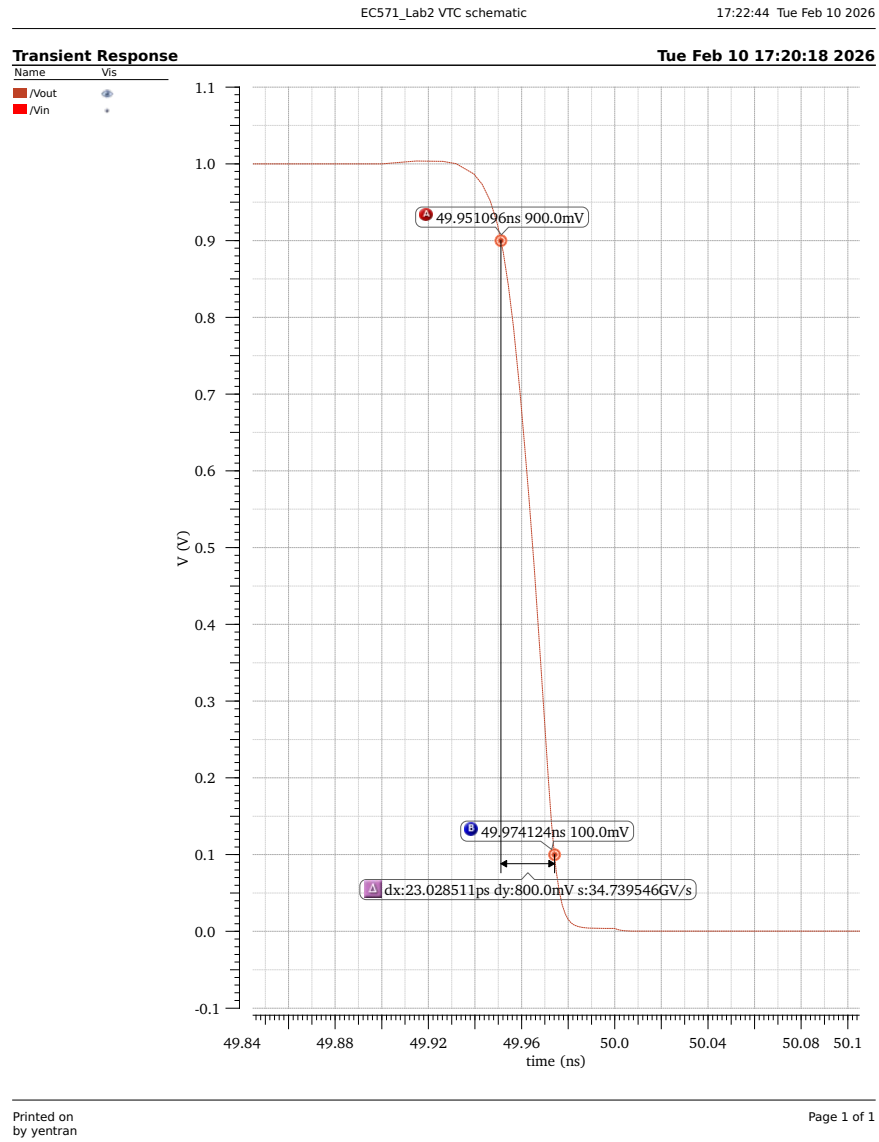


Figure 9: Fall time (t_f)

2.2 Part b

2.2.1 Propagation delay

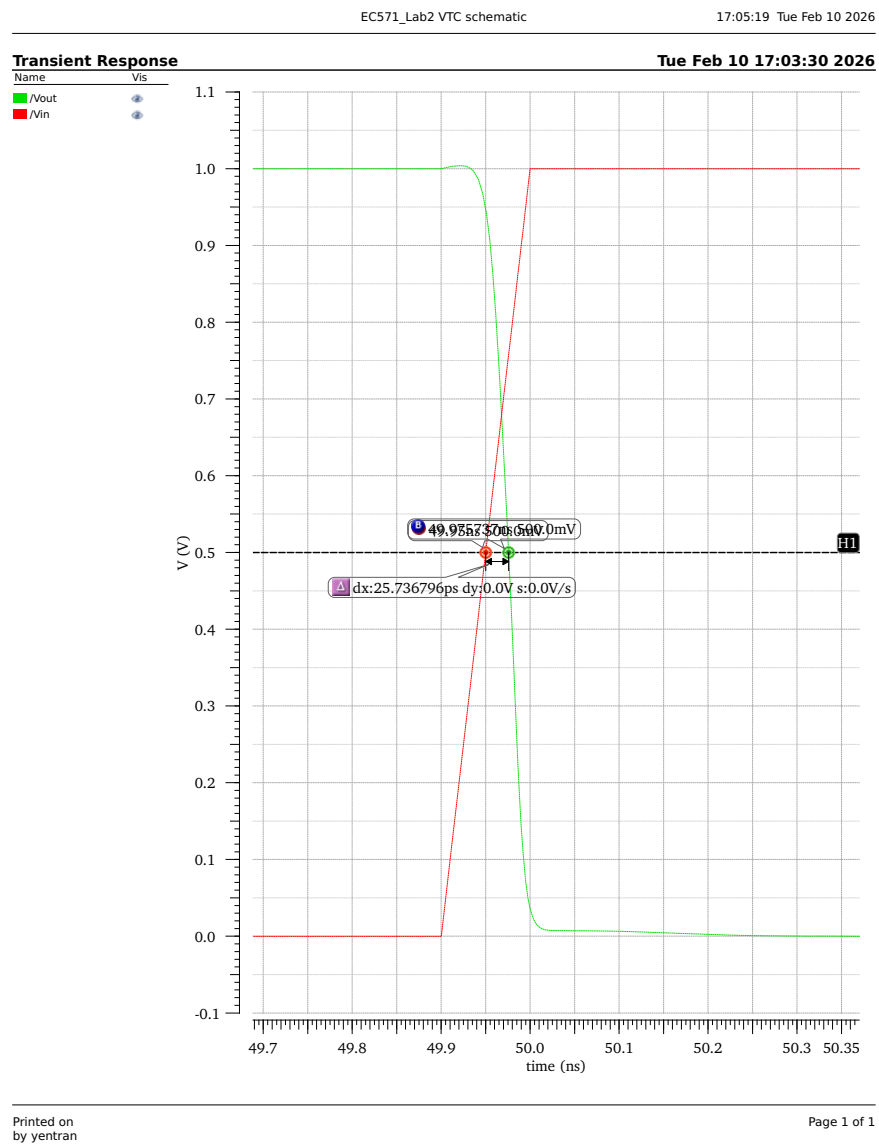


Figure 10: Propagation delay from High to Low (t_{pHL})

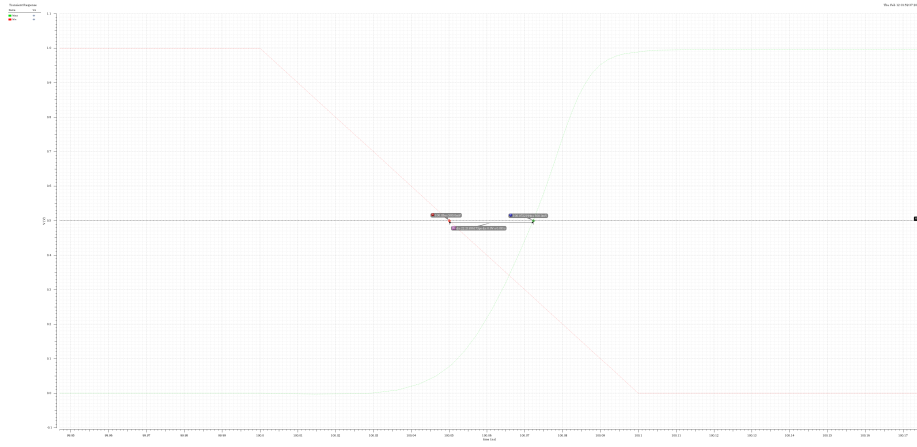


Figure 11: Propagation delay from Low to High (t_{pLH})

$$\text{Propagation delay}(t_p) = \frac{t_{pHL} + t_{pLH}}{2} = \frac{14.78 \text{ ps} + 11.19 \text{ ps}}{2} = 12.99 \text{ ps}$$

2.2.2 Rise time and Fall time

$$\text{Rise time } (t_r) = 100.086132 \text{ ns} - 100.052186 \text{ ns} = 33.946 \text{ ps}$$

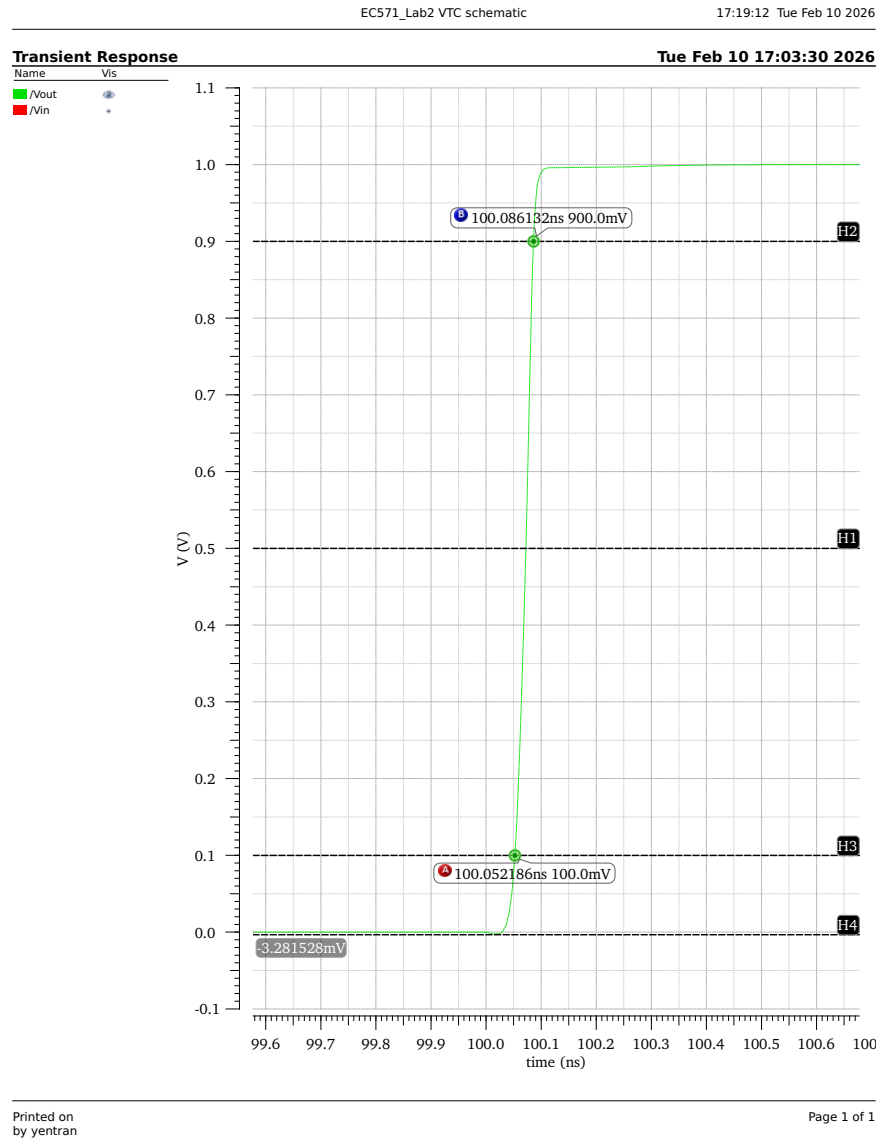


Figure 12: Rise time (t_r)

$$\text{Fall time } (t_f) = 49.99268 \text{ ns} - 49.95522 \text{ ns} = 37.46 \text{ ps}$$

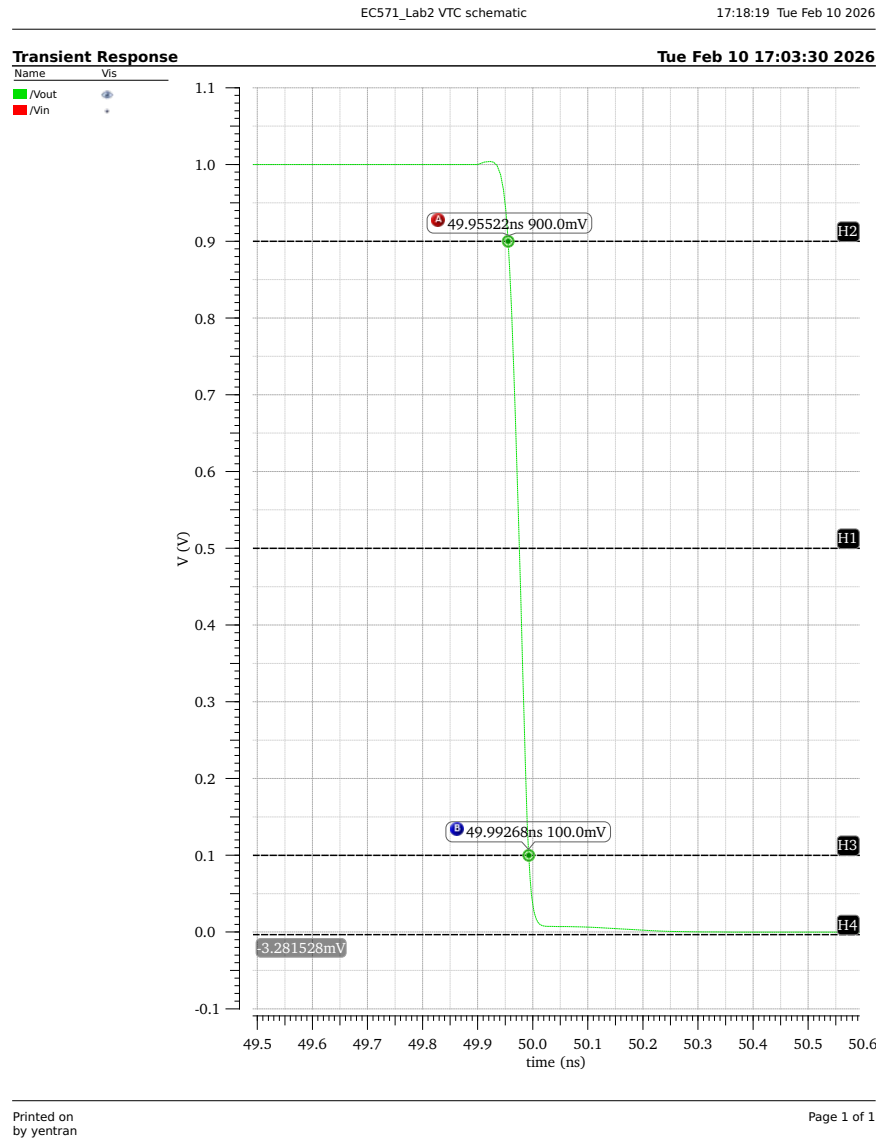


Figure 13: Fall time (t_f)

2.3 Part c

2.3.1 Propagation delay

$$t_{pHL} = 54.63 \text{ ps}$$

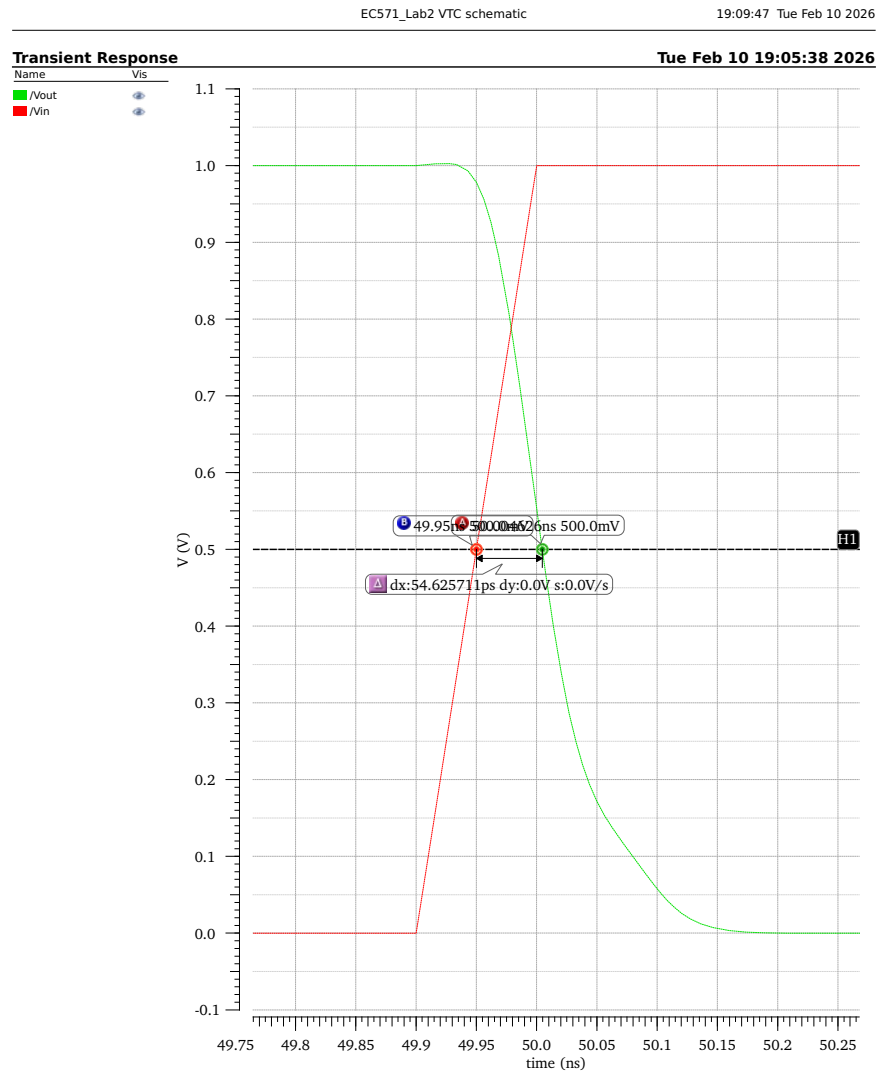


Figure 14: Propagation delay from High to Low (t_{pHL})

$$t_{pLH} = 49.46 \text{ ps}$$

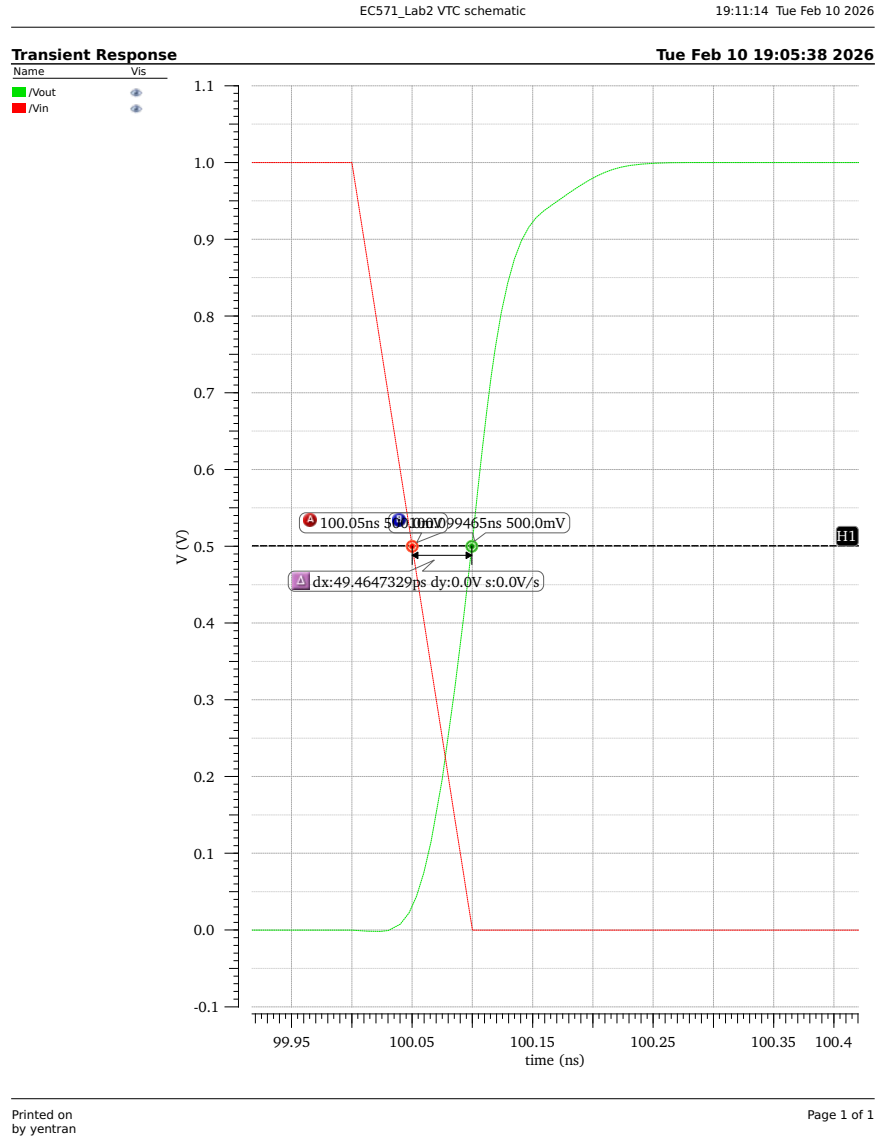


Figure 15: Propagation delay from Low to High (t_{pLH})

$$\text{Propagation delay}(t_p) = \frac{t_{pHL} + t_{pLH}}{2} = \frac{54.63 \text{ ps} + 49.46 \text{ ps}}{2} = 52.04 \text{ ps}$$

2.3.2 Rise time and Fall time

Rise time (t_r) = 77.92 ps

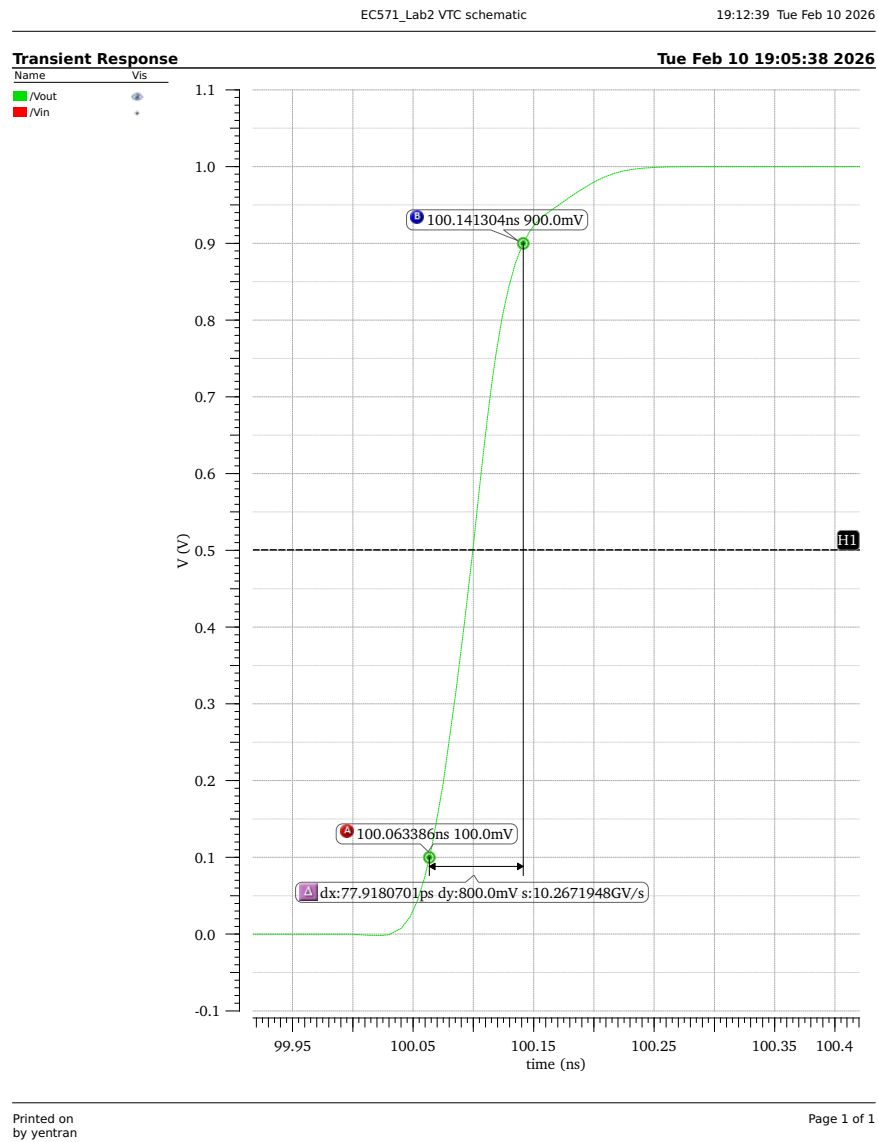


Figure 16: Rise time (t_r)

$$\text{Fall time } (t_f) = 113.93 \text{ ps}$$

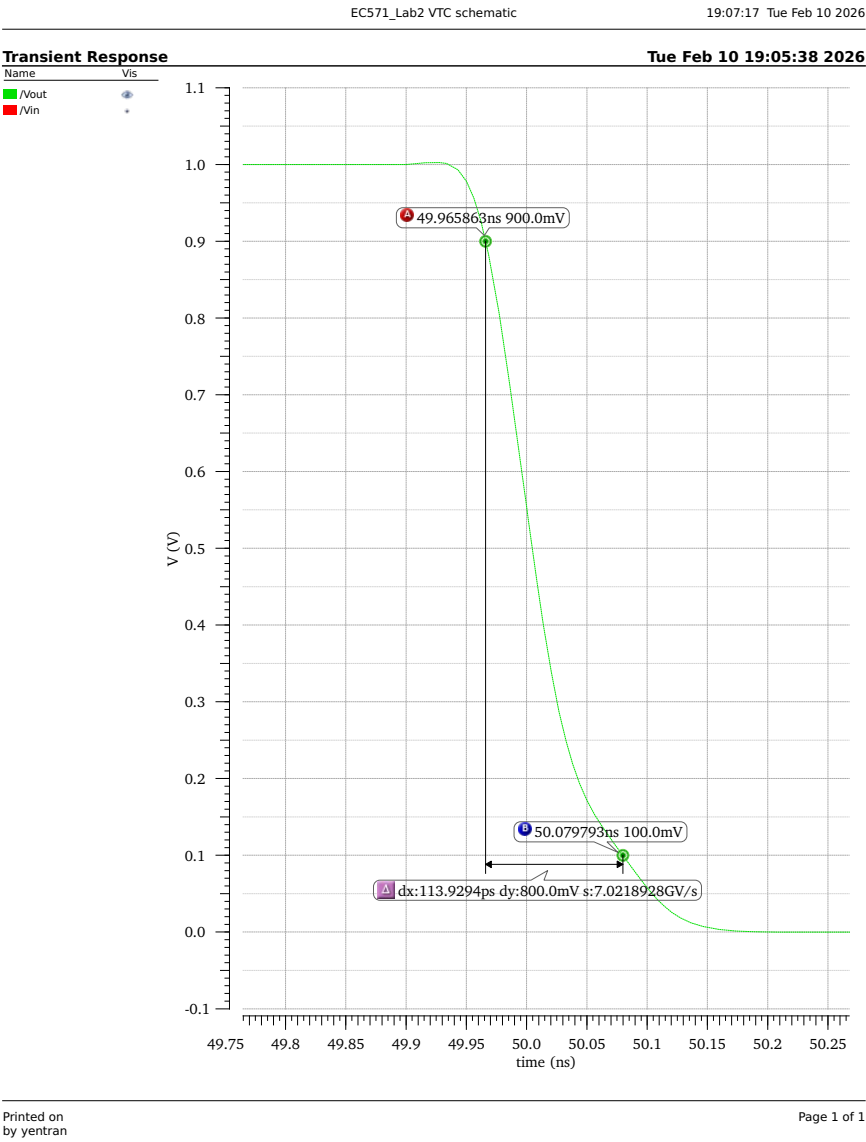


Figure 17: Fall time (t_f)

2.4 Part d

2.4.1 Propagation delay

$$t_{pHL} = 167.74 \text{ ps}$$

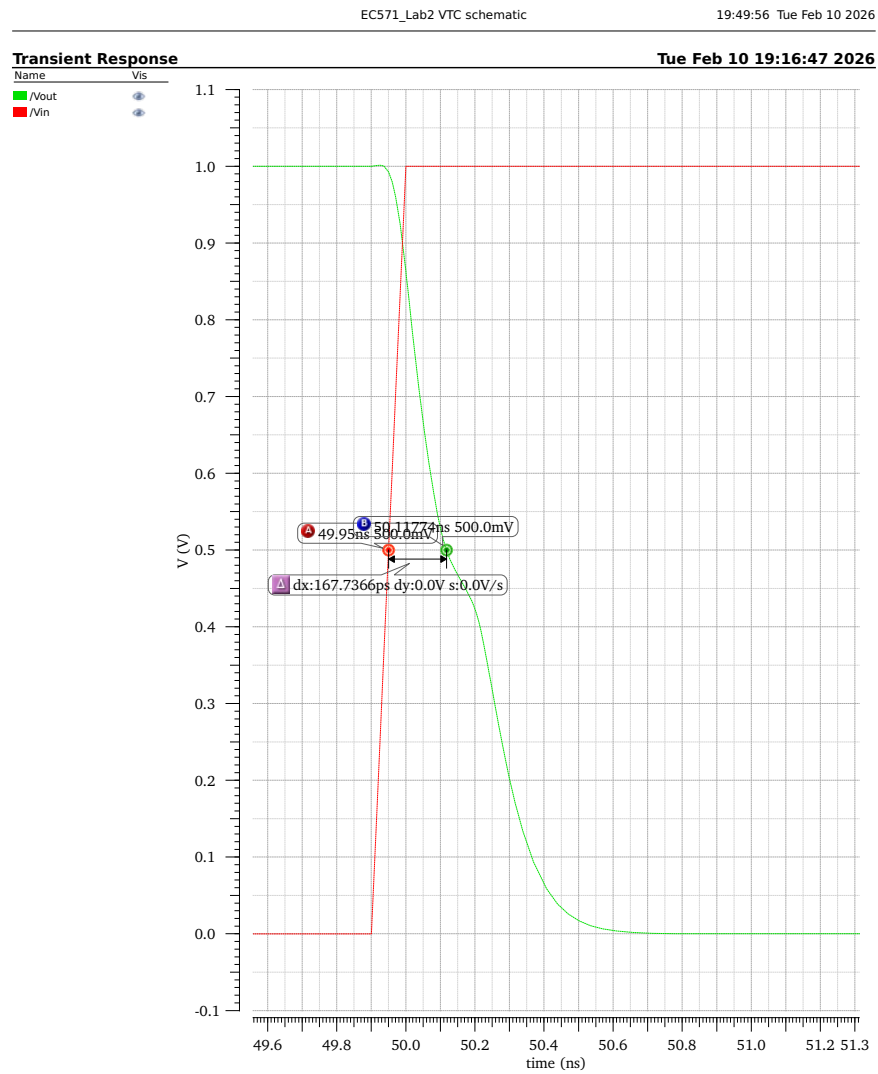


Figure 18: Propagation delay from High to Low (t_{pHL})

$$t_{pLH} = 139.87 \text{ ps}$$

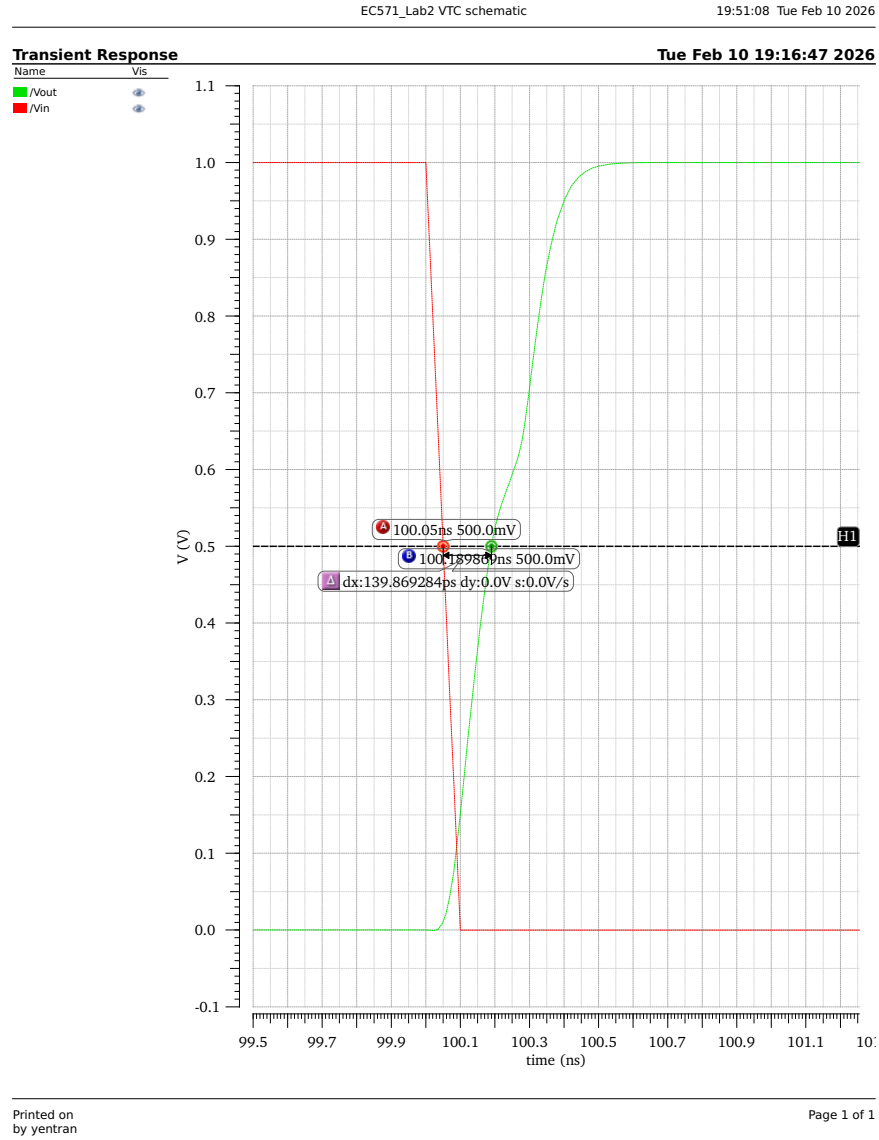


Figure 19: Propagation delay from Low to High (t_{pLH})

$$\text{Propagation delay}(t_p) = \frac{t_{pHL} + t_{pLH}}{2} = \frac{167.74 \text{ ps} + 139.87 \text{ ps}}{2} = 153.79 \text{ ps}$$

2.4.2 Rise time and Fall time

Rise time (t_r) = 279.53 ps

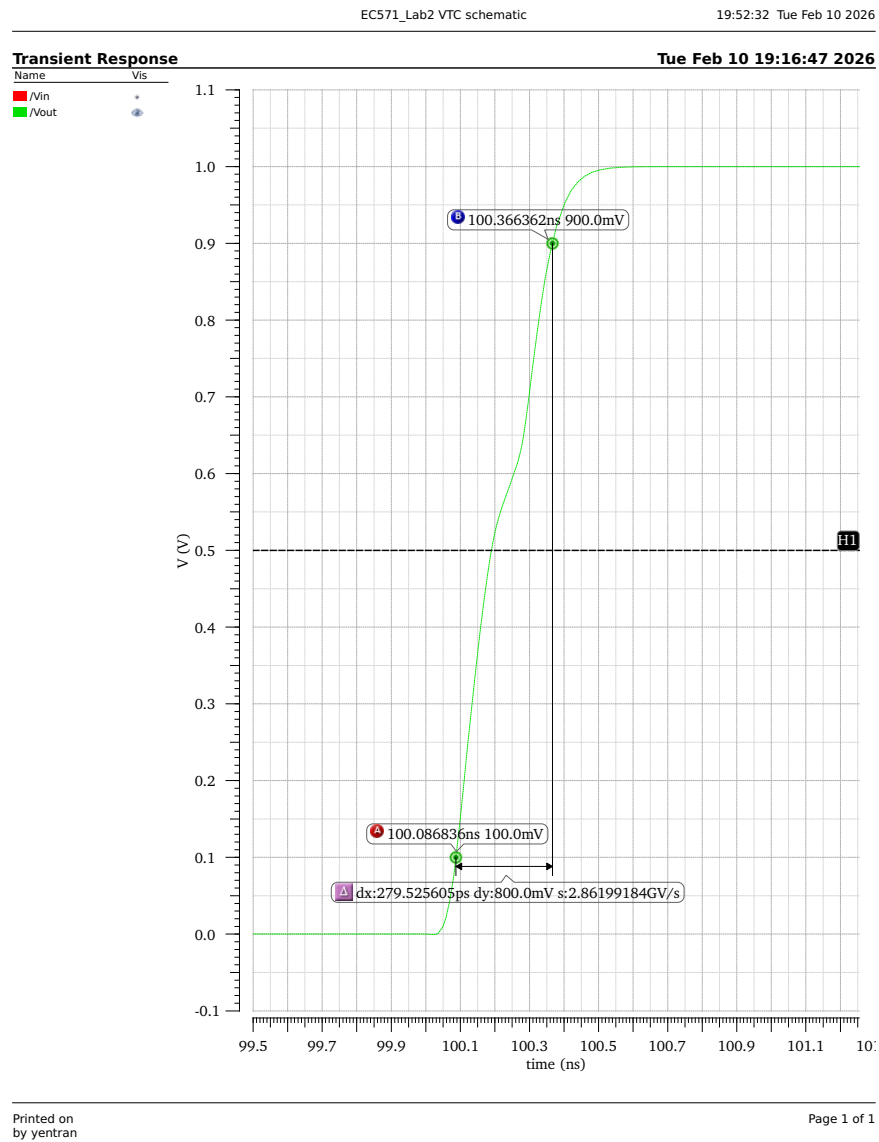


Figure 20: Rise time (t_r)

Fall time (t_f) = 374.85 ps

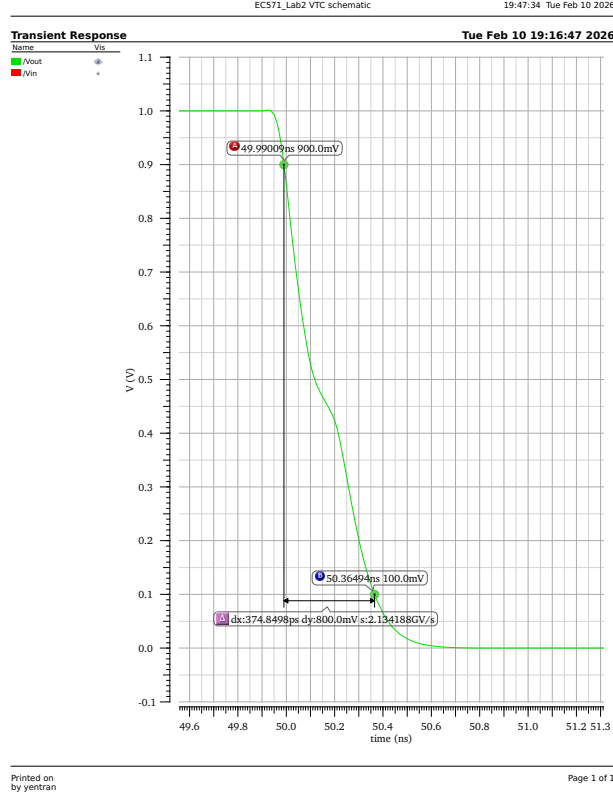


Figure 21: Fall time (t_f)

2.5 Part e

The larger the size of the second inverter compared to the first inverter, the higher the propagation delay. The fall time and rise time also increase as the inverter has to drive a larger size. This is because increasing the width will increase the capacitance, increasing the time the first inverter needed to charge to pull up or discharge to pull down. This increases the time propagation between input and output.

2.6 Part f

Part D: The margin noise at the input of the second inverter does not change between 4 implementations because the ratio between $\frac{w_p}{w_n} = 2$. Due $\frac{\mu_n}{\mu_p} = 2$, the ratio form a balance inverter.

3 Problem 3: Ring Oscillator

3.1 Part A

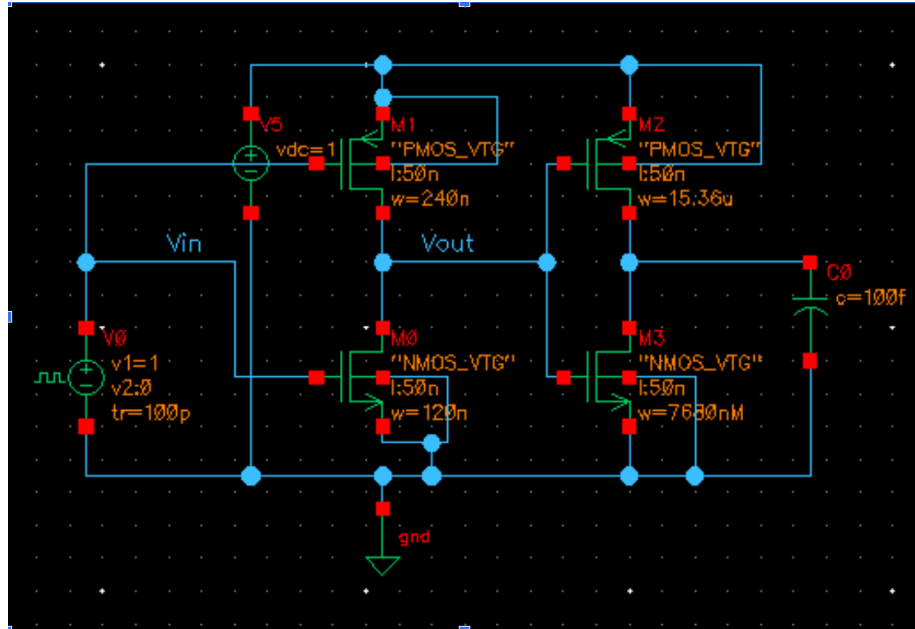


Figure 22: Ring Oscillator Schematic

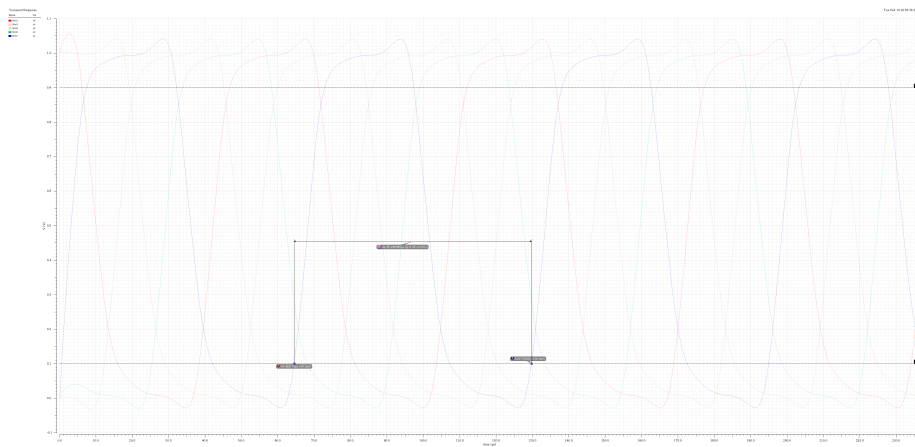


Figure 23: Ring Oscillator Schematic

3.2 Part B

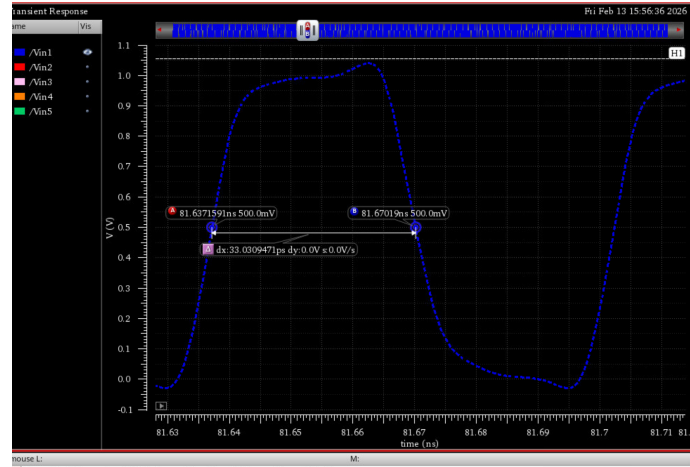


Figure 24: Ring Oscillator Schematic

3.3 Part C

The propagation through the Ring Oscillator passing through out 5 gates is half the period. 500 mV level mark is used to measure the time interval for $V_1 = 0$ to be turned back to $V_1 = 1$. The total time is 33.03 ps